

Brutal Series B02 - Nutrition in Animals | Acids, Bases & Salts | Fractions & Decimals | The Triangle & its Properties (~2x)

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. While chewing bread, the enzyme in saliva converts 18% of its starch to sugar every 30 seconds. At this steady rate, what fraction of the starch is converted in 50 seconds, and which enzyme does this? [\[Nutrition in](#)

[Animals\]](#)

- (A) 30%; trypsin
- (B) 30%; salivary amylase
- (C) 9%; salivary amylase
- (D) 18%; bile
- (E) 30%; pepsin
- (F) 60%; salivary amylase
- (G) 36%; salivary amylase

2. A bottle is $\frac{3}{4}$ full of juice. You pour out $\frac{8}{9}$ of the juice that is in it. What fraction of the FULL bottle's capacity is left in the bottle? [\[Fractions & Decimals\]](#)

- (A) $\frac{11}{12}$
- (B) $\frac{2}{3}$
- (C) $\frac{1}{9}$
- (D) $\frac{1}{4}$
- (E) $\frac{5}{12}$
- (F) $\frac{1}{12}$

3. In a tray of 20 test tubes, 8 hold acids and 7 hold bases; the rest hold neutral liquids. How many tubes are neutral, and which group turns RED litmus paper blue? [\[Acids, Bases & Salts\]](#)

- (A) 5; both acids and bases
- (B) 13; the bases
- (C) 8; the bases
- (D) 5; the bases
- (E) 5; the acids
- (F) 15; the bases
- (G) 5; the neutral ones

4. Two angles of a triangle are 47 deg and 58 deg. What is the size of the exterior angle at the third vertex?

[\[The Triangle & its Properties\]](#)

- (A) 75 deg
- (B) 122 deg
- (C) 115 deg
- (D) 105 deg
- (E) 285 deg
- (F) 133 deg
- (G) 95 deg

5. The small intestine of an adult is about 7.5 m long and the large intestine about 1.5 m. What percent of their combined length is the large intestine (to one decimal)? [\[Nutrition in Animals\]](#)
- (A) 1.5%
 - (B) 12%
 - (C) 18.8%
 - (D) 20%
 - (E) 83.3%
 - (F) 15%
 - (G) about 16.7%
6. A wire 2.5 m long is cut into pieces, each 0.4 m long. How many full pieces can you cut, and how much wire is left over? [\[Fractions & Decimals\]](#)
- (A) 6 pieces; 0.25 m left
 - (B) 5 pieces; 0.5 m left
 - (C) 6 pieces; 0.4 m left
 - (D) 10 pieces; 0.1 m left
 - (E) 7 pieces; 0.1 m left
 - (F) 6.25 pieces; nothing left
 - (G) 6 pieces; 0.1 m left
7. When an acid reacts with a base, the products are a salt and water. If hydrochloric acid reacts with sodium hydroxide, what salt forms, and what is this kind of reaction called? [\[Acids, Bases & Salts\]](#)
- (A) sodium chloride (common salt); neutralization
 - (B) sodium chloride; evaporation
 - (C) calcium carbonate; neutralization
 - (D) sodium hydroxide; neutralization
 - (E) carbon dioxide; neutralization
 - (F) sodium sulphate; neutralization
 - (G) sodium chloride; oxidation
8. An exterior angle of a triangle is 120 deg. One of its two opposite interior angles is 45 deg. What is the other opposite interior angle? [\[The Triangle & its Properties\]](#)
- (A) 35 deg
 - (B) 85 deg
 - (C) 120 deg
 - (D) 75 deg
 - (E) 105 deg
 - (F) 165 deg
9. A cow first swallows grass quickly, then later brings it back to chew slowly as 'cud'. If a cow spends 8 hours a day chewing cud, what fraction of the whole day (in lowest terms) is that, and what is this process called? [\[Nutrition in Animals\]](#)
- (A) $\frac{2}{3}$; rumination
 - (B) $\frac{1}{3}$; digestion
 - (C) $\frac{1}{4}$; rumination
 - (D) $\frac{8}{24}$; rumination
 - (E) $\frac{1}{3}$; rumination
 - (F) 3; rumination
 - (G) $\frac{1}{2}$; rumination

10. Which is the largest of these four numbers: 0.6, $\frac{3}{5}$, 0.59, $\frac{5}{8}$? [\[Fractions & Decimals\]](#)

- (A) they are all equal
- (B) $\frac{5}{8}$
- (C) 0.6 and $\frac{3}{5}$ are largest and equal
- (D) 0.59
- (E) 0.59
- (F) 0.6

11. An antacid is a mild base that cancels excess stomach acid. If 3 antacid tablets neutralize 15 mL of acid, how many tablets are needed for 40 mL of the same acid? [\[Acids, Bases & Salts\]](#)

- (A) 8 tablets
- (B) 120 tablets
- (C) 10 tablets
- (D) 6 tablets
- (E) 2.67 tablets
- (F) 13 tablets

12. In an isosceles triangle the vertex (apex) angle is 40 deg. What is the size of each equal base angle?

[\[The Triangle & its Properties\]](#)

- (A) 140 deg
- (B) 50 deg
- (C) 100 deg
- (D) 35 deg
- (E) 40 deg
- (F) 70 deg
- (G) 20 deg

13. An amoeba traps food using finger-like extensions and captures 3 food particles every 5 minutes. How many particles does it capture in 40 minutes, and what are these extensions called? [\[Nutrition in Animals\]](#)

- (A) 24; villi
- (B) 21; pseudopodia
- (C) 15; pseudopodia
- (D) 120; pseudopodia
- (E) 8; pseudopodia
- (F) 24; pseudopodia

14. A jug holds $\frac{7}{8}$ litre of milk. Each glass holds $\frac{7}{24}$ litre. How many glasses can be filled exactly from the jug? [\[Fractions & Decimals\]](#)

- (A) $\frac{8}{3}$ glasses
- (B) 21 glasses
- (C) $\frac{24}{8}$ glasses
- (D) 2 glasses
- (E) $\frac{49}{192}$ glasses
- (F) 3 glasses

15. A bee or red-ant sting injects an acid into the skin, causing pain. Rubbing which household item relieves it, and why does that work? [\[Acids, Bases & Salts\]](#)

- (A) vinegar, because it is an acid
- (B) more acid, to balance it
- (C) baking soda, because it is a mild base that neutralizes the acid
- (D) water alone, because it dilutes the base
- (E) lemon juice, because it is an acid
- (F) common salt, because it is neutral

16. Each base angle of an isosceles triangle is 65 deg. What is the size of its vertex angle? [\[The Triangle & its Properties\]](#)

- (A) 30 deg
- (B) 25 deg
- (C) 45 deg
- (D) 50 deg
- (E) 65 deg
- (F) 130 deg
- (G) 115 deg

17. The inner wall of the small intestine is folded into tiny finger-like projections that multiply the absorbing surface. If a smooth patch of 50 cm² is increased 30 times by these projections, what is the new absorbing area, and what are the projections called? [\[Nutrition in Animals\]](#)

- (A) 500 cm²; villi
- (B) 1530 cm²; villi
- (C) 80 cm²; villi
- (D) 150 cm²; villi
- (E) 1500 cm²; cilia
- (F) 1500 cm²; villi
- (G) 15 cm²; villi

18. A ribbon is 5 m long. You cut as many $\frac{2}{3}$ -m pieces as possible. How many whole pieces do you get, and what length of ribbon is left over? [\[Fractions & Decimals\]](#)

- (A) 3 pieces; $\frac{1}{3}$ m left
- (B) 7 pieces; $\frac{2}{3}$ m left
- (C) 7 pieces; $\frac{1}{2}$ m left
- (D) 7 pieces; $\frac{1}{3}$ m left
- (E) 10 pieces; $\frac{1}{3}$ m left
- (F) 7.5 pieces; none left
- (G) 6 pieces; $\frac{1}{3}$ m left

19. A yellow turmeric stain on cloth turns red where soap is rubbed on it, then turns yellow again when a drop of lemon juice is added. What does this tell you about soap, and what does the lemon do? [\[Acids, Bases & Salts\]](#)

- (A) soap is acidic; the lemon makes it more basic
- (B) soap is basic; the lemon (acid) neutralizes the base and restores yellow
- (C) soap is basic; the lemon is also basic
- (D) soap is basic; the lemon adds new dye
- (E) soap is a salt; the lemon dissolves it
- (F) soap is acidic; the lemon is also acidic
- (G) soap is neutral; the lemon adds colour

20. An equilateral triangle has a perimeter of 7.5 cm. What is the size of each angle, and the length of each side? [\[The Triangle & its Properties\]](#)

- (A) 180 deg; 2.5 cm
- (B) 90 deg; 2.5 cm
- (C) 60 deg; 2.5 cm
- (D) 60 deg; 7.5 cm
- (E) 60 deg; 3.75 cm
- (F) 60 deg; 2.25 cm
- (G) 45 deg; 1.5 cm

21. An adult has 32 teeth: 8 incisors, 4 canines, 8 premolars and 12 molars. What fraction of the teeth are canines (in lowest terms), and what is a canine's main job? [\[Nutrition in Animals\]](#)

- (A) $\frac{1}{4}$; tearing
- (B) $\frac{4}{32}$; tearing
- (C) $\frac{1}{8}$; tearing
- (D) $\frac{3}{8}$; tearing
- (E) $\frac{1}{2}$; tearing
- (F) $\frac{1}{8}$; grinding
- (G) $\frac{1}{8}$; cutting

22. Work out $(1\frac{1}{2} + 2\frac{1}{4}) \times \frac{2}{3}$. [\[Fractions & Decimals\]](#)

- (A) $1\frac{1}{2}$
- (B) $6\frac{1}{4}$
- (C) $4\frac{1}{2}$
- (D) $\frac{30}{12}$
- (E) $\frac{2}{3}$
- (F) $3\frac{3}{4}$
- (G) $2\frac{1}{2}$

23. Different natural acids come from different foods. Which acids are found in vinegar and in curd, respectively? [\[Acids, Bases & Salts\]](#)

- (A) acetic acid; oxalic acid
- (B) acetic acid; lactic acid
- (C) citric acid; tartaric acid
- (D) lactic acid; acetic acid
- (E) tartaric acid; acetic acid
- (F) formic acid; lactic acid

24. A right-angled triangle has legs of 6 cm and 8 cm. How long is its hypotenuse? [\[The Triangle & its Properties\]](#)

- (A) 7 cm
- (B) 12 cm
- (C) 48 cm
- (D) 100 cm
- (E) 14 cm
- (F) 10 cm
- (G) 28 cm

25. The liver makes bile, which is stored in the gall bladder before being released into the small intestine. If the liver makes 800 mL of bile a day and the gall bladder holds 50 mL, how many times the gall-bladder volume is made each day, and what does bile do to fat? [\[Nutrition in Animals\]](#)

- (A) 16 times; digests fat completely
- (B) 0.0625 times; breaks fat into tiny droplets
- (C) 16 times; breaks fat into tiny droplets
- (D) 16 times; digests protein
- (E) 8 times; breaks fat into tiny droplets
- (F) 750 times; breaks fat into tiny droplets
- (G) 40 times; breaks fat into tiny droplets

26. Eight identical coins together weigh exactly 1 kg. How many such coins are there in 3.5 kg? [\[Fractions & Decimals\]](#)

- (A) 32 coins
- (B) 4 coins
- (C) 280 coins
- (D) 0.4375 coins
- (E) 28 coins
- (F) 24 coins
- (G) 27 coins

27. A farmer finds his soil is too acidic for crops to grow well. Which substance should he add to fix it, and what is the chemistry behind the fix? [\[Acids, Bases & Salts\]](#)

- (A) more acidic fertilizer, to balance it
- (B) salt, because it is neutral
- (C) vinegar, because it adds acid
- (D) sugar, because plants like it
- (E) lime (a base), because it neutralizes the extra acid
- (F) lemon juice, because it is sour
- (G) sand, because it is dry

28. In a right-angled triangle the hypotenuse is 13 cm and one leg is 5 cm. How long is the other leg? [\[The Triangle & its Properties\]](#)

- (A) 8 cm
- (B) 6 cm
- (C) 12 cm
- (D) 9 cm
- (E) 13.9 cm
- (F) 144 cm
- (G) 18 cm

29. A meal contains 40 g of protein. About 25% of it is broken down in the stomach and the rest in the small intestine. How many grams are digested in the small intestine, and which juice there finishes protein digestion? [\[Nutrition in Animals\]](#)

- (A) 30 g; saliva
- (B) 15 g; pancreatic and intestinal juice
- (C) 30 g; gastric juice
- (D) 30 g; pancreatic and intestinal juice
- (E) 10 g; pancreatic and intestinal juice
- (F) 26 g; pancreatic and intestinal juice
- (G) 8 g; pancreatic and intestinal juice

30. In a class, $\frac{2}{3}$ of the students are boys, and $\frac{3}{4}$ of the boys play cricket. What fraction of the whole class are boys who play cricket? [\[Fractions & Decimals\]](#)

- (A) $\frac{3}{8}$
- (B) $\frac{1}{4}$
- (C) $\frac{5}{12}$
- (D) $\frac{17}{12}$
- (E) $\frac{6}{7}$
- (F) $\frac{2}{4}$
- (G) $\frac{1}{2}$

31. A factory must neutralize 500 L of acidic waste before releasing it, using 2 g of lime per litre. How much lime is needed in total, and why must this be done? [\[Acids, Bases & Salts\]](#)

- (A) 100 g; to make it acidic
- (B) 2 kg; to neutralize the acid
- (C) 500 g; to neutralize the acid
- (D) 250 g; to neutralize the acid
- (E) 1000 kg; to neutralize the acid
- (F) 1 kg; to neutralize the acid so it does not harm river life

32. A 17 m ladder leans against a wall with its foot 8 m from the wall. How high up the wall does the ladder reach? [\[The Triangle & its Properties\]](#)

- (A) 18.8 m
- (B) 12 m
- (C) 23 m
- (D) 11 m
- (E) 9 m
- (F) 15 m

33. A cow's gut bacteria help it turn grass into useful matter. If 200 kg of grass eaten gives 30 kg of dung, what percent of the grass leaves as dung? [\[Nutrition in Animals\]](#)

- (A) 15 kg
- (B) 30%
- (C) 6.67%
- (D) 170%
- (E) 15%
- (F) 85%
- (G) 0.15%

34. From a 10 m roll of cloth, three pieces of 3.75 m, 2.6 m and 1.85 m are cut. How much cloth is left?

[Fractions & Decimals]

- (A) 1.2 m
- (B) 1.8 m
- (C) 8.2 m
- (D) 2.2 m
- (E) 0.18 m
- (F) 4.2 m
- (G) 2.8 m

35. A drop of china rose (gudhal) extract is added to an unknown liquid and turns GREEN. Is the liquid acidic, basic or neutral, and what colour would it have turned in an acid? [Acids, Bases & Salts]

- (A) neutral; it would have stayed green
- (B) neutral; it would have turned pink in an acid
- (C) basic; it would have turned dark pink in an acid
- (D) acidic; it would have turned green in an acid
- (E) basic; it would have turned blue in an acid
- (F) basic; it would have stayed green in an acid
- (G) acidic; it would have turned pink in an acid

36. Can a triangle be made with sides 3 cm, 4 cm and 8 cm? Why? [The Triangle & its Properties]

- (A) Yes, because $3 + 4 + 8 = 15$
- (B) Yes, any three lengths form a triangle
- (C) No, because all sides are different
- (D) No, because $3 + 8$ is less than 4
- (E) No, because $4 + 8$ is less than 3
- (F) Yes, because 8 is the longest side
- (G) No, because $3 + 4$ is less than 8

37. A child has 20 milk teeth, while an adult has 32 permanent teeth. How many MORE teeth does the adult set have, and what is special about permanent teeth? [Nutrition in Animals]

- (A) 12 more; they grow back like milk teeth
- (B) 12 more; they do not grow back if lost
- (C) 640 more; they do not grow back if lost
- (D) 8 more; they do not grow back if lost
- (E) 12 more; they fall out by age 12
- (F) 52 more; they do not grow back if lost
- (G) 12 fewer; they do not grow back if lost

38. A snail crawls $\frac{3}{5}$ of a metre each minute. How far does it travel in one quarter of an hour? [Fractions &

Decimals]

- (A) 2.4 m
- (B) 7.5 m
- (C) 12 m
- (D) 9 m
- (E) 0.15 m
- (F) 45 m

- 39.** Sweet-loving bacteria in the mouth turn sugar into an acid that attacks tooth enamel. Brushing with a basic toothpaste protects teeth. What process lets the toothpaste protect the teeth? [\[Acids, Bases & Salts\]](#)
- (A) filtration of the sugar
 - (B) condensation of the acid
 - (C) neutralization of the acid by the basic toothpaste
 - (D) oxidation of the enamel
 - (E) the toothpaste adding more acid
 - (F) freezing of the acid
 - (G) evaporation of the acid
- 40.** The three angles of a triangle are in the ratio 1 : 2 : 3. Find the value of the smallest angle and classify the triangle. [\[The Triangle & its Properties\]](#)
- (A) 30 deg; it is right-angled
 - (B) 60 deg; it is equilateral
 - (C) 30 deg; it is acute
 - (D) 36 deg; it is right-angled
 - (E) 45 deg; it is right-angled
 - (F) 30 deg; it is obtuse
 - (G) 20 deg; it is acute
- 41.** A 500 kcal meal is eaten and the small intestine absorbs about 90% of its energy-giving nutrients. How many kilocalories are absorbed, and through which structures? [\[Nutrition in Animals\]](#)
- (A) 410 kcal; villi
 - (B) 450 kcal; the stomach wall
 - (C) 45 kcal; villi
 - (D) 490 kcal; villi
 - (E) 450 kcal; villi
 - (F) 550 kcal; villi
- 42.** A 6.3 m ribbon is cut into 0.9 m pieces. Each piece is sold for Rs 4. How many pieces are there, and how much money is collected? [\[Fractions & Decimals\]](#)
- (A) 0.7 pieces; Rs 2.80
 - (B) 8 pieces; Rs 32
 - (C) 7 pieces; Rs 32
 - (D) 70 pieces; Rs 280
 - (E) 7 pieces; Rs 28
 - (F) 7 pieces; Rs 24
- 43.** Which natural indicator stays yellow in acidic and neutral liquids but turns red only when added to a base? [\[Acids, Bases & Salts\]](#)
- (A) litmus
 - (B) phenol red
 - (C) tea leaves
 - (D) red cabbage
 - (E) beetroot juice
 - (F) turmeric
 - (G) china rose

44. What is the sum of all three exterior angles of any triangle (one at each vertex)? [\[The Triangle & its Properties\]](#)

- (A) 720 deg
- (B) 360 deg
- (C) 90 deg
- (D) 540 deg
- (E) 180 deg
- (F) 300 deg

45. A cow needs 25 kg of fodder each day. A farmer has 875 kg of fodder stored. For how many full days will it last, and in which stomach chamber is the swallowed grass first stored? [\[Nutrition in Animals\]](#)

- (A) 850 days; the rumen
- (B) 35 days; the gizzard
- (C) 40 days; the rumen
- (D) 35 days; the stomach acid pouch
- (E) 30 days; the rumen
- (F) 35 days; the rumen
- (G) 3.5 days; the rumen

46. Which is greater, $\frac{5}{6}$ or $\frac{7}{9}$, and by how much? [\[Fractions & Decimals\]](#)

- (A) $\frac{7}{9}$ is greater, by $\frac{2}{15}$
- (B) $\frac{7}{9}$ is greater, by $\frac{1}{18}$
- (C) $\frac{5}{6}$ is greater, by $\frac{1}{2}$
- (D) $\frac{5}{6}$ is greater, by $\frac{1}{18}$
- (E) $\frac{5}{6}$ is greater, by $\frac{1}{3}$
- (F) they are equal

47. Adding baking soda (a base) to vinegar (an acid) fizzes strongly. Besides a salt and water, which gas is released in this neutralization? [\[Acids, Bases & Salts\]](#)

- (A) chlorine
- (B) water vapour
- (C) hydrogen
- (D) nitrogen
- (E) oxygen
- (F) ammonia
- (G) carbon dioxide

48. How many medians does a triangle have, and at what single point do they all meet? [\[The Triangle & its Properties\]](#)

- (A) 6; the orthocentre
- (B) 3; the vertex
- (C) 1; the centroid
- (D) 3; the midpoint
- (E) 3; the incentre
- (F) 3; the centroid
- (G) 2; the centroid

49. Humans make about 1.5 litres of saliva each day. Over a 30-day month, how many litres is that, and one job of saliva (besides moistening food) is to begin digesting which nutrient? [\[Nutrition in Animals\]](#)

- (A) 1.5 L; starch
- (B) 45 L; protein
- (C) 31.5 L; starch
- (D) 4.5 L; starch
- (E) 45 L; fat
- (F) 45 L; starch
- (G) 450 L; starch

50. A cake recipe needs $2\frac{1}{4}$ cups of flour per cake. How many cups are needed for 3 cakes? [\[Fractions & Decimals\]](#)

- (A) $7\frac{1}{2}$ cups
- (B) $27/4$ cups (unsimplified form expected)
- (C) 8 cups
- (D) $6\frac{1}{4}$ cups
- (E) $6\frac{1}{2}$ cups
- (F) $6\frac{3}{4}$ cups
- (G) $5\frac{1}{4}$ cups

51. Fresh milk slowly turns sour as it stands. Why does it turn sour, and why does adding a pinch of baking soda delay the souring? [\[Acids, Bases & Salts\]](#)

- (A) lactic acid forms; baking soda freezes it
- (B) acetic acid forms; baking soda dilutes it
- (C) lactic acid forms; baking soda is a base that neutralizes the acid
- (D) lactic acid forms; baking soda adds more acid
- (E) salt forms; baking soda dissolves it
- (F) citric acid forms; baking soda evaporates it
- (G) sugar forms; baking soda sweetens it

52. In a right-angled triangle, two of the three altitudes happen to be sides of the triangle. Which sides are they? [\[The Triangle & its Properties\]](#)

- (A) all three sides
- (B) the base only
- (C) the hypotenuse and one leg
- (D) the hypotenuse only
- (E) the two legs (the sides forming the right angle)
- (F) none of the sides
- (G) the two medians

53. The stomach lining is protected from its own acid by a slimy layer. If a healthy 5 mm layer of this lining is reduced by 40% due to illness, how thick is it now, and what is the protective layer made of? [\[Nutrition in Animals\]](#)

- (A) 1.5 mm; mucus
- (B) 7 mm; mucus
- (C) 2 mm; mucus
- (D) 3 mm; mucus
- (E) 4.4 mm; mucus
- (F) 3 mm; bile
- (G) 3.4 mm; mucus

54. A jar is 0.375 full of water. Express this as a fraction in its lowest terms. [\[Fractions & Decimals\]](#)

- (A) $\frac{3}{80}$
- (B) $\frac{3}{8}$
- (C) $\frac{1}{3}$
- (D) $\frac{375}{1000}$
- (E) $\frac{37}{100}$
- (F) $\frac{3}{4}$
- (G) $\frac{5}{8}$

55. Limewater turns milky when carbon dioxide is passed through it. Limewater is a base; what is its chemical name? [\[Acids, Bases & Salts\]](#)

- (A) calcium carbonate
- (B) calcium sulphate
- (C) sodium carbonate
- (D) calcium hydroxide
- (E) sodium hydroxide
- (F) calcium chloride
- (G) calcium oxide

56. A right-angled triangle has legs of 9 cm and 12 cm. What is its area? [\[The Triangle & its Properties\]](#)

- (A) 45 cm²
- (B) 67.5 cm²
- (C) 54 cm²
- (D) 108 cm²
- (E) 60 cm²
- (F) 27 cm²
- (G) 36 cm²

57. A liver makes about 0.6 L of bile per day. How much bile does it make in one week, and into which organ is this bile released to act on fats? [\[Nutrition in Animals\]](#)

- (A) 4.2 L; the small intestine
- (B) 4.2 L; the stomach
- (C) 4.2 L; the large intestine
- (D) 3.6 L; the small intestine
- (E) 0.6 L; the small intestine
- (F) 42 L; the small intestine

58. Evaluate using the correct order of operations: $\frac{1}{2} + \frac{1}{2} \div \frac{1}{2}$. [\[Fractions & Decimals\]](#)

- (A) $\frac{1}{2}$
- (B) 2
- (C) $1\frac{1}{4}$
- (D) $1\frac{1}{2}$
- (E) $\frac{3}{4}$
- (F) 1
- (G) $2\frac{1}{2}$

59. A student tests 12 household items: 5 are acidic, 4 are basic, and the rest are neutral. How many are neutral, and which group includes soap and baking soda? [\[Acids, Bases & Salts\]](#)

- (A) 9; the basic group
- (B) 4; the basic group
- (C) 3; the neutral group
- (D) 3; the acidic group
- (E) 3; the basic group
- (F) 1; the basic group
- (G) 7; the basic group

60. Two angles of a triangle are in the ratio 2 : 3, and the third angle is 80 deg. What is the smallest angle of the triangle? [\[The Triangle & its Properties\]](#)

- (A) 30 deg
- (B) 32 deg
- (C) 40 deg
- (D) 20 deg
- (E) 50 deg
- (F) 60 deg

61. About 2.5 L of water enters the large intestine each day, and roughly 90% is reabsorbed. How much leaves the body as watery waste, and what is the large intestine's main job? [\[Nutrition in Animals\]](#)

- (A) 500 mL; absorbing water
- (B) 250 mL; absorbing water
- (C) 2250 mL; absorbing water
- (D) 250 mL; digesting food
- (E) 2.25 L; absorbing water
- (F) 90 mL; absorbing water
- (G) 25 mL; absorbing water

62. A car uses 6.5 litres of petrol for every 100 km. How much petrol does it use for a 250 km journey?

[\[Fractions & Decimals\]](#)

- (A) 16.5 L
- (B) 16.25 L
- (C) 15.25 L
- (D) 13 L
- (E) 1625 L
- (F) 26 L
- (G) 162.5 L

- 63.** In a titration-style test, 30 mL of an acid is exactly neutralized by 20 mL of a base. What is the simplest acid-to-base ratio, and how much base would neutralize 90 mL of the same acid? [\[Acids, Bases & Salts\]](#)
- (A) 2 : 3; 60 mL
 - (B) 3 : 2; 135 mL
 - (C) 3 : 2; 30 mL
 - (D) 1 : 1; 90 mL
 - (E) 3 : 2; 60 mL
 - (F) 3 : 2; 45 mL
 - (G) 2 : 3; 45 mL
- 64.** An exterior angle of a triangle is 120 deg, and its two opposite interior angles are in the ratio 1 : 3. What are these two interior angles? [\[The Triangle & its Properties\]](#)
- (A) 20 deg and 100 deg
 - (B) 45 deg and 75 deg
 - (C) 30 deg and 90 deg
 - (D) 40 deg and 80 deg
 - (E) 30 deg and 60 deg
 - (F) 24 deg and 96 deg
 - (G) 60 deg and 60 deg
- 65.** Grass is mostly cellulose, which humans cannot digest. Cows manage it because their rumen holds helpful microbes. Why can a cow digest grass while a human cannot? [\[Nutrition in Animals\]](#)
- (A) Cows have more teeth than humans
 - (B) Humans cannot swallow grass at all
 - (C) Cow saliva contains amylase that digests cellulose
 - (D) Cows chew grass into sugar with their teeth
 - (E) Grass turns to fat in the cow's liver
 - (F) Cows have gut microbes that release cellulose-digesting enzymes; humans lack them
 - (G) Humans have no stomach acid to digest grass
- 66.** A person spends $\frac{1}{3}$ of their income on rent and $\frac{1}{4}$ on food, and saves the rest. If the savings are Rs 5000, what is the total income? [\[Fractions & Decimals\]](#)
- (A) Rs 20000
 - (B) Rs 7000
 - (C) Rs 12000
 - (D) Rs 60000
 - (E) Rs 5000
 - (F) Rs 9000
- 67.** Acid rain slowly damages marble monuments such as the Taj Mahal. Acid rain harms marble because the rain is what, and the gas that makes rain acidic comes mainly from burning what? [\[Acids, Bases & Salts\]](#)
- (A) acidic; oxygen from plants
 - (B) basic; clean firewood
 - (C) acidic; fuels that release sulphur and nitrogen gases
 - (D) basic; factory smoke that is alkaline
 - (E) neutral; pure water vapour
 - (F) acidic; carbon-free fuels

- 68.** A right-angled triangle is also isosceles (its two legs are equal). What is the size of each of its two equal angles? [\[The Triangle & its Properties\]](#)
- (A) 55 deg
 - (B) 22.5 deg
 - (C) 60 deg
 - (D) 45 deg
 - (E) 50 deg
 - (F) 90 deg
- 69.** In an amoeba, captured food is digested inside a small bubble of fluid. If an amoeba forms 2 such bubbles every hour, how many form in 6 hours, and what is each bubble called? [\[Nutrition in Animals\]](#)
- (A) 12; a contractile vacuole
 - (B) 18; a food vacuole
 - (C) 3; a food vacuole
 - (D) 12; a food vacuole
 - (E) 12; a chloroplast
 - (F) 12; a food vacuole is wrong... a cell wall
 - (G) 6; a food vacuole
- 70.** Calculate $0.2 \times 0.5 \times 0.5$. [\[Fractions & Decimals\]](#)
- (A) 0.005
 - (B) 0.25
 - (C) 0.1
 - (D) 1.2
 - (E) 5
 - (F) 0.05
- 71.** Your stomach makes an acid that helps digest food and kills germs. Too much of it causes 'acidity', cured by a base. Which acid is present in the stomach? [\[Acids, Bases & Salts\]](#)
- (A) sulphuric acid
 - (B) lactic acid
 - (C) hydrochloric acid
 - (D) nitric acid
 - (E) citric acid
 - (F) acetic acid
 - (G) carbonic acid
- 72.** A triangle has sides 6 cm, 6 cm and 9 cm. How is it classified by its sides, and how do its two base angles compare? [\[The Triangle & its Properties\]](#)
- (A) scalene; two angles are equal
 - (B) equilateral; all angles are equal
 - (C) isosceles; the two base angles are different
 - (D) equilateral; the base angles differ
 - (E) right-angled; angles 90, 45, 45
 - (F) isosceles; the two base angles are equal
 - (G) scalene; all angles are different

73. Among a person's 32 teeth there are 8 incisors and 4 canines. What is the simplest ratio of incisors to canines? [\[Nutrition in Animals\]](#)
- (A) 2 : 3
 - (B) 4 : 1
 - (C) 1 : 2
 - (D) 2 : 1
 - (E) 3 : 1
 - (F) 8 : 4
74. A tank is $\frac{4}{5}$ full. Then an amount equal to $\frac{1}{4}$ of the FULL tank is drained out. What fraction of the tank is full now? [\[Fractions & Decimals\]](#)
- (A) $\frac{11}{20}$
 - (B) $\frac{1}{20}$
 - (C) $\frac{21}{20}$
 - (D) $\frac{16}{20}$
 - (E) $\frac{9}{20}$
 - (F) $\frac{7}{20}$
 - (G) $\frac{3}{5}$
75. A student dips turmeric paper in soap solution and it turns red. She then adds lemon juice drop by drop until the paper turns yellow again. What single process has she demonstrated? [\[Acids, Bases & Salts\]](#)
- (A) filtration
 - (B) sublimation
 - (C) condensation
 - (D) neutralization
 - (E) sedimentation
 - (F) evaporation
76. In a right-angled triangle the two legs are 21 cm and 28 cm. How long is the hypotenuse? [\[The Triangle & its Properties\]](#)
- (A) 24.5 cm
 - (B) 1225 cm
 - (C) 7 cm
 - (D) 42 cm
 - (E) 33 cm
 - (F) 35 cm
77. The acidic food leaving the stomach is made non-acidic in the small intestine by bile and pancreatic juice so that enzymes can work. As it moves from the stomach to the small intestine, the food changes from acidic to what? [\[Nutrition in Animals\]](#)
- (A) unchanged acidic
 - (B) oily and basic
 - (C) basic (alkaline)
 - (D) sweet
 - (E) neutral salt only
 - (F) even more acidic
 - (G) gas

- 78.** Shop A sells 5 pens for Rs 12.50; Shop B sells 8 pens for Rs 18. Which shop is cheaper per pen, and by how much? [\[Fractions & Decimals\]](#)
- (A) Shop B; cheaper by Rs 0.25 per pen
 - (B) Shop A; cheaper by Rs 0.50 per pen
 - (C) Shop A; cheaper by Rs 0.25 per pen
 - (D) Shop A; cheaper by Rs 6 per pen
 - (E) both cost the same per pen
 - (F) Shop B; cheaper by Rs 5.50 per pen
 - (G) Shop B; cheaper by Rs 0.50 per pen
- 79.** A wasp sting is basic, unlike a bee sting which is acidic. Which household item should be used to treat a wasp sting, and why? [\[Acids, Bases & Salts\]](#)
- (A) baking soda (a base), because it neutralizes the sting
 - (B) sugar, because it soothes
 - (C) plain water, because it is acidic
 - (D) salt, because it is neutral
 - (E) lemon, because it is also basic
 - (F) vinegar (a mild acid), because it neutralizes the basic sting
- 80.** In a triangle, angle A is twice angle B, and angle C is 20 deg more than angle B. What is the largest angle? [\[The Triangle & its Properties\]](#)
- (A) 70 deg
 - (B) 120 deg
 - (C) 80 deg
 - (D) 40 deg
 - (E) 60 deg
 - (F) 90 deg
- 81.** Of 30 g of fat eaten, the special lymph vessels in the villi absorb about 95%. How many grams of fat are absorbed this way, and what are these vessels called? [\[Nutrition in Animals\]](#)
- (A) 25 g; lacteals
 - (B) 28.5 g; lacteals
 - (C) 30 g; lacteals
 - (D) 285 g; lacteals
 - (E) 1.5 g; lacteals
 - (F) 2.85 g; lacteals
 - (G) 28.5 g; capillaries
- 82.** Add the fractions $\frac{3}{8} + \frac{5}{12}$. [\[Fractions & Decimals\]](#)
- (A) $\frac{19}{20}$
 - (B) $\frac{2}{5}$
 - (C) $\frac{15}{96}$
 - (D) $\frac{11}{24}$
 - (E) $\frac{19}{24}$
 - (F) $\frac{8}{20}$
 - (G) $\frac{1}{2}$

- 83.** A bottle holds 50 antacid tablets. If a patient takes 2 tablets after lunch and 2 after dinner every day, for how many full days will the bottle last, and how many tablets are left over? [\[Acids, Bases & Salts\]](#)
- (A) 12 days; 0 left
 - (B) 10 days; 10 left
 - (C) 12.5 days; 0 left
 - (D) 12 days; 2 tablets left
 - (E) 13 days; none left
 - (F) 25 days; 0 left
- 84.** Two sides of a triangle are 7 cm and 10 cm. Between which two values must the third side lie? [\[The Triangle & its Properties\]](#)
- (A) exactly 3 cm or 17 cm
 - (B) between 10 cm and 17 cm
 - (C) between 7 cm and 10 cm
 - (D) between 0 cm and 17 cm
 - (E) between 3 cm and 17 cm
 - (F) between 3 cm and 10 cm
 - (G) between 17 cm and 70 cm
- 85.** Food travels through a long tube from mouth to anus. Which organ does food reach immediately AFTER the stomach, and where does most absorption of nutrients take place? [\[Nutrition in Animals\]](#)
- (A) the small intestine; the stomach
 - (B) the oesophagus; the stomach
 - (C) the rectum; the large intestine
 - (D) the liver; the small intestine
 - (E) the large intestine; the small intestine
 - (F) the stomach again; the large intestine
 - (G) the small intestine; the small intestine
- 86.** How many $\frac{3}{4}$ -kg packets can be filled from 4 $\frac{1}{2}$ kg of sugar? [\[Fractions & Decimals\]](#)
- (A) 12 packets
 - (B) 4 packets
 - (C) 6 packets
 - (D) 5 packets
 - (E) $\frac{36}{6}$ packets
 - (F) $\frac{1}{6}$ packet
- 87.** A nettle plant stings because its hairs inject the same acid as a red ant. What is the name of this acid, and what soothes the sting? [\[Acids, Bases & Salts\]](#)
- (A) formic acid; more acid
 - (B) acetic acid; an acid such as vinegar
 - (C) tartaric acid; salt
 - (D) formic acid; vinegar
 - (E) citric acid; baking soda
 - (F) lactic acid; lemon juice
 - (G) formic (methanoic) acid; a mild base such as baking soda

88. A television's screen has a diagonal of 40 inches and a width of 32 inches. What is its height? [\[The Triangle](#)

[& its Properties\]](#)

- (A) 18 inches
- (B) 36 inches
- (C) 8 inches
- (D) 24 inches
- (E) 30 inches
- (F) 51 inches
- (G) 28 inches

89. A person chews each mouthful of food 40 times, and eats 25 mouthfuls in a meal. How many chewing actions is that in the meal, and what kind of digestion is chewing? [\[Nutrition in Animals\]](#)

- (A) 100; mechanical digestion
- (B) 320; mechanical digestion
- (C) 640; mechanical digestion
- (D) 1000; chemical digestion
- (E) 65; mechanical digestion
- (F) 1000; mechanical (physical) digestion
- (G) 1000; absorption

90. A worker paints $\frac{2}{5}$ of a wall on the first day and $\frac{1}{3}$ of the wall on the second day. What fraction of the wall is still unpainted? [\[Fractions & Decimals\]](#)

- (A) $\frac{11}{15}$
- (B) $\frac{4}{15}$
- (C) $\frac{2}{8}$
- (D) $\frac{3}{8}$
- (E) $\frac{13}{15}$
- (F) $\frac{1}{15}$

91. Common table salt is made from a strong acid and a strong base. When dissolved in water, how does this salt affect litmus paper? [\[Acids, Bases & Salts\]](#)

- (A) it turns both litmus papers red
- (B) it turns blue litmus green
- (C) it does not change either litmus (it is neutral)
- (D) it turns blue litmus red
- (E) it turns both litmus papers blue
- (F) it turns red litmus blue

92. A median of a triangle is drawn from one vertex. Into two triangles of equal WHAT does it divide the original triangle? [\[The Triangle & its Properties\]](#)

- (A) equal shape
- (B) equal sides
- (C) equal area
- (D) equal height
- (E) equal perimeter
- (F) equal hypotenuse
- (G) equal angles

93. Carbohydrate digestion ends mainly as a simple sugar that the body can absorb. After a starchy meal is fully digested, what is the main end product, and where is it absorbed into the blood? [\[Nutrition in Animals\]](#)

- (A) glucose; the mouth
- (B) glucose; the large intestine
- (C) amino acids; the stomach
- (D) glucose; the small intestine
- (E) fatty acids; the small intestine
- (F) starch; the small intestine

94. Three-quarters of a number is 0.6 more than one-half of the same number. What is the number?

[\[Fractions & Decimals\]](#)

- (A) 4.8
- (B) 0.3
- (C) 6
- (D) 1.2
- (E) 2.4
- (F) 0.8
- (G) 0.15

95. In a neutralization, 4 g of acid reacts completely with 6 g of base to give 9 g of salt. Using conservation of mass, how much water is also formed? [\[Acids, Bases & Salts\]](#)

- (A) 1 kg
- (B) 19 g
- (C) 1 g
- (D) 0 g
- (E) 9 g
- (F) 10 g
- (G) 2 g

96. A right-angled triangle has legs of 8 cm and 15 cm. What is its perimeter? [\[The Triangle & its Properties\]](#)

- (A) 120 cm
- (B) 38 cm
- (C) 23 cm
- (D) 17 cm
- (E) 60 cm
- (F) 40 cm
- (G) 33 cm

97. Place the steps of nutrition in order. After food is absorbed, which step lets body cells actually USE the nutrients, and which final step removes undigested waste? [\[Nutrition in Animals\]](#)

- (A) ingestion; digestion
- (B) digestion; absorption
- (C) assimilation; egestion
- (D) digestion; egestion
- (E) absorption; assimilation
- (F) assimilation; ingestion
- (G) egestion; assimilation

98. A recipe needs $\frac{3}{4}$ kg of flour. A cook has already added 0.35 kg, then adds another $\frac{1}{5}$ kg. How many more kilograms of flour are still needed? [\[Fractions & Decimals\]](#)

- (A) 0.05 kg
- (B) 0.4 kg
- (C) 0.25 kg
- (D) 0.1 kg
- (E) 0.55 kg
- (F) 0.95 kg
- (G) 0.2 kg

99. A gardener's soil is too acidic at pH 4. Adding lime raises the pH by 0.5 for every 200 g used. How many grams of lime are needed to reach the neutral pH of 7, and is the soil then acidic, basic or neutral?

[\[Acids, Bases & Salts\]](#)

- (A) 1200 g; neutral
- (B) 1200 g; basic
- (C) 3000 g; neutral
- (D) 1200 g; acidic
- (E) 2400 g; neutral
- (F) 1400 g; neutral

100. In a triangle, the exterior angle at one vertex is 130 degrees, and the two remote interior angles are in the ratio 3:2. What is the SMALLER of those two interior angles? [\[The Triangle & its Properties\]](#)

- (A) 78 deg
- (B) 65 deg
- (C) 52 deg
- (D) 26 deg
- (E) 104 deg
- (F) 48 deg
- (G) 50 deg